What actually moves virtual screening performance: two measured resolution limits and a pre-registered decomposition of six interventions

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Abstract

Reported improvements to structure-based virtual screening are usually differences in AUROC or enrichment between two protocols on one benchmark. We asked what a benchmark of typical size can actually resolve, and then measured six interventions against that limit with the reading rule for each committed to version control before the data existed.

Two limits, not one. Scoring the *same* Vina poses with a second implementation of the same scoring function — agreeing at r = 0.970, no compound differing by more than 2 kcal/mol — moves AUROC by 0.020 [95% CI 0.011, 0.029]. Re-running the whole protocol at a different search effort and scoring it the same way moves AUROC by 0.039 [0.024, 0.056]. The first bounds a comparison of scoring functions on fixed poses; the second bounds a comparison of protocols, which is what most published improvements are. Both fall inside the range in which docking improvements are typically reported.

On SARS-CoV-2 Mpro: changing the scoring function on identical poses moved AUROC +0.140, repairing a receptor preparation defect +0.045, scoring the pose ensemble rather than the top pose +0.055, substituting DiffDock-L pose generation +0.017, supplying the correct binding site to a co-folding model −0.013, and eight-fold search effort −0.019. Two clear the relevant floor, one is marginal, three fall inside. These are distributions over cross-validation fold assignment, not point estimates: we previously published the pose-ensemble arm as +0.019 and refuted, and it proved to be the 2nd percentile of a distribution centred at +0.031. The search-effort result is the sharpest either way: its measured effect is smaller than the run-to-run reproducibility of the protocol it was measured in.

Docking adds nothing demonstrable over seven free physicochemical descriptors on either panel we assembled: on Mpro it significantly subtracts (−0.002 [−0.004, −0.000]) and on Factor Xa its +0.015 [−0.003, +0.034] crosses zero. Applying the same descriptors to LIT-PCBA gives a median AUROC of 0.7260 across 15 targets against published AutoDock Vina at 0.581 and GNINA at 0.611–0.616. These descriptor baselines are supervised on each benchmark's own labels while docking is not, so this is a bias control in the sense of Sieg et al., not a claim that descriptors are a deployable substitute. On the AVE-debiased variant the descriptor median falls to 0.6510.

One method exceeded the floor. Boltz-2, an open-weights co-folding model scoring compounds from sequence and SMILES alone, does not beat the descriptor baseline on its own (Mpro +0.026, Factor Xa +0.019, both intervals spanning zero) but adds +0.093 [+0.062, +0.125] and +0.075 [+0.051, +0.099] respectively when combined with those descriptors. Its advantage is not that it is less property-driven — descriptors explain 63% of its score and 63% of Vina's — but that on Mpro its residual, the part orthogonal to those descriptors, ranks actives at 0.657 where six docking scores residualised the same way sat between 0.494 and 0.573. Both descriptor comparisons are supervised on each target's own labels; transferred cold to the other target the descriptor model ranks actives *below chance*, and the standalone Boltz-2 score — which requires no labels at all — is the only configuration usable on a target with no known actives. Across twelve cross-validation fold seeds the two do not overlap: the highest docking residual observed is 0.581, the lowest Boltz-2 residual 0.644.

1. Introduction

A virtual screening method is normally shown to work by reporting enrichment on a retrospective benchmark. Two properties of that evidence are rarely reported alongside it. The first is what difference the benchmark could distinguish from noise. The second is which component of a multi-part protocol produced the difference — absent that, a pipeline change gets attributed to whichever part its authors were working on.

Analogue and decoy bias in retrospective sets is well documented [1,2], and LIT-PCBA [3] was constructed to control it — though LIT-PCBA has itself since been audited and found to carry leakage and redundancy across its splits [8]. That literature establishes that reported enrichment can measure property matching. It does not establish how much of a *given* reported improvement is resolvable at all, which is the question here.

This paper reports a resolution limit, measured rather than assumed and in two variants; a set of six single-factor comparisons against it, each with its reading rule fixed in advance; and a descriptor baseline on LIT-PCBA, to test whether our own panels are unrepresentative.

2. Methods

Benchmarks. Two panels assembled from ChEMBL with measured actives and measured inactives: SARS-CoV-2 Mpro (PDB 7VU6, 751 compounds, 34.2% active) and Factor Xa (886 compounds, 45.7% active; an earlier 854-compound freeze of the same panel is used for the marginal-value comparison in §3.4 and is labelled where quoted). LIT-PCBA [3] full and AVE-debiased variants, 15 targets, used as distributed.

Descriptor baseline. Molecular weight, clogP, hydrogen-bond donors, hydrogen-bond acceptors, rotatable bonds, topological polar surface area, formal charge. Logistic regression, out-of-fold predictions under 5-fold stratified cross-validation. On the AVE variant the authors' own train/validation split is honoured rather than re-folded, so the AVE and full-set figures differ in estimator as well as in compound selection. Inactives capped at 25,000 per LIT-PCBA target by seeded random sampling.

Docking. AutoDock Vina 1.2.5, 22 Å box, receptor passing a redocking control at the screening protocol. Exhaustiveness was 4 for the scoring-function, receptor-repair and pose-ensemble arms; the search-effort arm compares exhaustiveness 4 against 32. gnina 1.3 CNN rescoring applied to the identical Vina output poses. DiffDock-L for the pose-source arm, with poses minimised before scoring. Supplementary Table S2 gives the receptor state, exhaustiveness and baseline for each arm.

Boltz-2. Version 2.2.1, protein sequence plus ligand SMILES, `affinity_probability_binary` as the endpoint, fixed in advance. `diffusion_samples = 1` for cost. Pocket conditioning, where applied, supplied as 1-based sequence indices derived from residues within 8 Å of the docking box centre.

Pre-registration. Every reading rule — endpoint, threshold, control, abort condition — was committed to a public git repository before the corresponding data existed. Supplementary Table S1 lists each arm's pre-registration file, commit hash and commit date. Amendments are appended rather than edited, with the reason recorded. Two comparisons were not pre-registered and are identified as such in S1.

Statistics. Paired bootstrap over compounds, 95% percentile intervals: 10,000 resamples for the interventions in §3.2 and for the resolution floors, 4,000 for the Boltz-2 contrasts. A difference is reported as demonstrated only when it exceeds the applicable resolution floor *and* its interval excludes zero.

3. Results

3.1 Two resolution limits, 0.020 and 0.039 AUROC

gnina's `--score_only` mode re-evaluates the Vina scoring function on a pose file it is handed. Applied to the exact poses Vina wrote at exhaustiveness 4, it reproduces the cached Vina score at r = 0.9700, mean difference −0.108 kcal/mol, with no compound differing by more than 2 kcal/mol. The two score vectors nonetheless give AUROC 0.3992 and 0.3791 — **0.020 apart [+0.011, +0.029]**. Call this the **scoring floor**: what two implementations of one scoring function differ by when placement is held fixed.

Comparing instead against the exhaustiveness-32 cache — same compounds, same scoring function, an independent stochastic search — gives r = 0.9372 and AUROC 0.4186 against 0.3793, **0.039 apart [+0.024, +0.056]**. Call this the **protocol floor**: run-to-run reproducibility of the pipeline end to end.

Neither pair is a meaningfully different description of binding, yet the metric separates each by an amount comparable to published improvements. We report a difference as unresolvable when it falls below the floor that matches the comparison being made: the scoring floor where poses are held fixed, the protocol floor where placement or search is allowed to vary.

3.2 One intervention clears its floor; four fall inside it

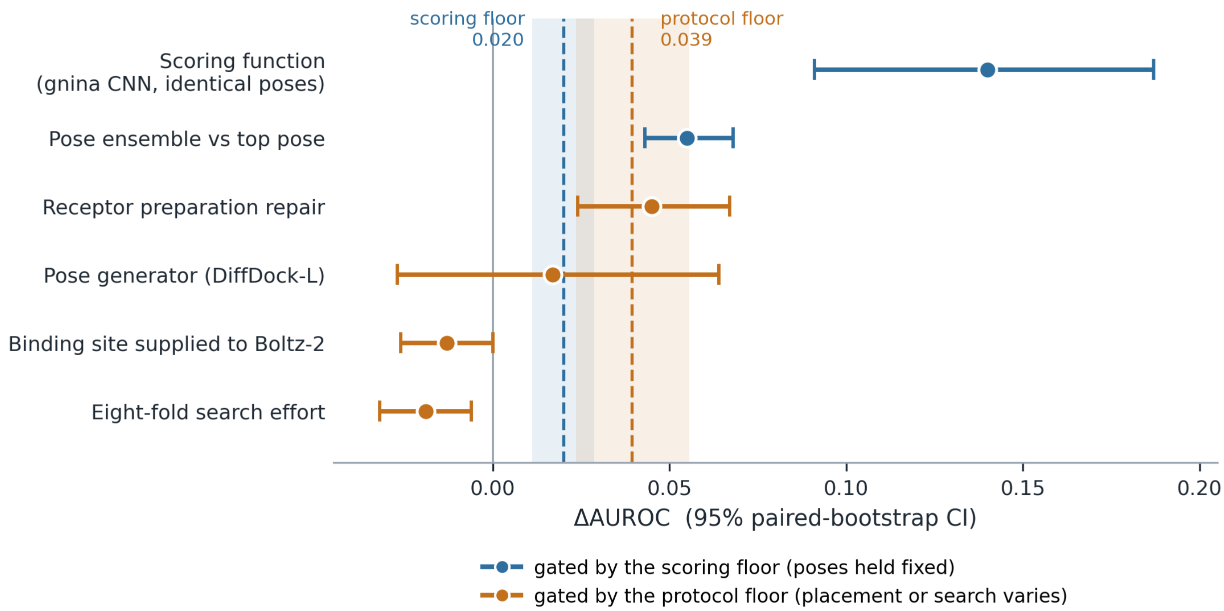

Figure 1: **Six interventions against the two measured resolution limits.** Points are ΔAUROC with 95% paired-bootstrap intervals; dashed lines are the two floors from §3.1 with their own bootstrap intervals shaded. Each intervention is coloured by the floor that applies to it — the scoring floor where poses were held fixed, the protocol floor where placement or search was allowed to vary. The two floor intervals overlap between 0.024 and 0.029. Note that the eight-fold search-effort comparison *is* the measurement that defines the protocol floor, so its effect cannot exceed it by construction.

intervention	ΔAUROC	95% CI	pre-registered bar	pre-registered verdict
Scoring function, identical poses (gnina CNN)	**+0.140**	[+0.091, +0.187]	CI entirely above +0.10	directionally supported, **not at full strictness** (lower bound 0.091)
Receptor preparation repair	**+0.045**	[+0.024, +0.067]	CI entirely above 0	**supported**
Pose ensemble rather than top pose	**+0.055**	[+0.043, +0.068]	> +0.04, CI excluding 0	**supported** — corrects a previously published refutation, see below
Pose generator substituted (DiffDock-L)	+0.017	[−0.027, +0.064]	CI above +0.05 would refute "scoring is the problem"	scoring hypothesis **retained**
Binding site supplied to Boltz-2	−0.013	[−0.026, +0.000]	> +0.04, CI excluding 0	**not demonstrated**
Eight-fold search effort	−0.019	[−0.032, −0.006]	*not pre-registered*	—

On re-gating, and why we do not do it silently. Four of these bars were fixed before the resolution limits in §3.1 were separated, and three of them cite a single "≈0.04 measurement floor" that we now know conflates the scoring and protocol cases. A pre-registration cannot be moved after the fact, so the verdicts above are read against the bars as written. The corrected floors are reported alongside as a secondary observation, and they change the picture in one place only:

intervention	poses fixed?	applicable floor	reading against it
Scoring function (gnina CNN)	yes	scoring, 0.020	clears
Receptor preparation repair	receptor only; ligands re-docked	protocol, 0.039	**marginal** — the point estimate exceeds the floor but the two intervals overlap across nearly their whole width
Pose ensemble	yes	scoring, 0.020	**clears** — all 40 fold seeds above it, lowest +0.040
Pose generator (DiffDock-L)	no	protocol, 0.039	inside
Binding site to Boltz-2	n/a — different model	protocol, 0.039	inside
Eight-fold search effort	no	protocol, 0.039	inside

The receptor-repair row is the one place the two readings diverge: it passes its pre-registered bar cleanly and is only marginal against a floor fixed afterwards. We report both and claim neither over the other.

These are six single-factor comparisons, not a partition of one pipeline. Four were run on the donor-defective receptor described below, before it was repaired; the receptor-repair row measures what that condition costs, and Supplementary Table S2 gives each arm's baseline in full. Their nulls are therefore conditional on a receptor now known to be broken.

Two interventions clear their floor, and they do not divide as cleanly as *what is computed* versus *where the ligand is placed*. Changing the scoring function clears by a wide margin; so does scoring the pose ensemble rather than the top pose — a placement-side intervention, in that it reads the distribution the search already produced. What does **not** matter is generating the poses differently, or generating more of them. The narrower statement the data supports is that search *effort* and pose *source* are inert, while how the pose distribution is *read* is not.

The search-effort row makes the point most economically: its CI excludes zero, so the effect is real and negative, but it is smaller than the protocol floor — and the exhaustiveness-4/32 pair *is* the protocol floor, so eight times the compute is by construction indistinguishable from running the same protocol twice. The pose-source result is next sharpest: Vina and DiffDock poses correlate at r = 0.139, nearly uncorrelated coordinates, and produce rankings differing by less than the floor.

The pose-ensemble arm carries a correction. We previously published this arm at +0.019 with a pre-registered verdict of *refuted*, measured on the donor-defective receptor. Re-run on the repaired receptor across 40 cross-validation fold seeds it gives **+0.055 [+0.043, +0.068]**, every seed above the floor. The earlier figure is not a contradictory result: on the defective receptor the same pipeline gives +0.031 with a full seed range of [+0.017, +0.043], and the published +0.019 sits at its 2nd percentile. That refutation was a single draw from a distribution straddling the floor, and the honest verdict on that data was always *unresolved*. Paired per seed, repairing the receptor contributes +0.024 [+0.011, +0.039] to this arm and helps in 39 of 40 seeds.

The receptor defect is worth reporting for its own sake. Every receptor we had prepared carried zero hydrogen-bond donors: the converter typed all 754 Mpro nitrogens as acceptors and added no polar hydrogens, so the scoring function could not form a protein-donor hydrogen bond at all. Repairing it moved AUROC +0.045 and changed six of the top ten shortlisted compounds. A defect invisible in every summary statistic reordered most of the answer.

3.3 Seven descriptors reach 0.726 on LIT-PCBA, against published docking at 0.58–0.62

	median AUROC	targets
descriptors, LIT-PCBA full	**0.7260**	15
descriptors, full, ≥50 actives	0.6830	10
descriptors, AVE-debiased	**0.6510**	15
AutoDock Vina, published [5]	0.581	15
GNINA, published [5]	0.611–0.616	15

Per-target values are in Supplementary Table S4. Both published figures come from one source, Sunseri & Koes [5], Table 2; we are not aware of an independent second measurement of Vina on this benchmark. An independent 2026 benchmarking effort on LIT-PCBA [4] reports ROC-AUC “hovering near 0.6” for its strongest classical scorer, but its docking arm is AutoDock-GPU rather than Vina, so it is not a substitute for the Vina figure. On the full set, seven descriptors costing microseconds exceed published Vina by 0.145, or 0.102 restricting to the ten targets with ≥50 actives. Our own panels’ descriptor baselines (Mpro 0.7654, Factor Xa 0.7041) fall inside LIT-PCBA’s 0.577–0.864 range, so they are not unrepresentative.

What this comparison is and is not. Our descriptor baselines are supervised on each target’s own actives and inactives; docking scores require no labels. The comparison is therefore a bias control in the sense of Sieg et al. [2] — it bounds how much of a benchmark’s apparent enrichment is reachable from ligand properties alone — not a claim that descriptors are a deployable substitute for docking on a novel target. Sunseri & Koes [5] reach the opposite conclusion, that their CNN models significantly outperform simple-descriptor models; their descriptor models were fit off-benchmark to PDBbind and CrossDocked affinity data and transferred, and the difference in conclusion is a difference in what the baseline is allowed to see.

What was not matched. Our baselines use up to 25,000 inactives per target drawn by seeded random sampling, against published figures computed on the full inactive sets and, in [5], taking the maximum score over all reference receptors for the 13 of 15 targets with multiple templates. AUROC is insensitive to negative-class size in expectation, so we expect no systematic direction to the sampling difference, but this is not a matched comparison and we do not claim one.

AVE debiasing reduces the descriptor median to 0.6510, robust to excluding targets with few validation actives (0.6510, 0.6497, 0.6510 at ≥0, ≥20, ≥50 actives). Two things change between 0.7260 and 0.6510 — the compound selection and, because the AVE variant ships its own train/validation split, the estimator — so the 0.075 drop is attributable to the AVE variant as a whole rather than to debiasing alone. What survives is that 0.651 from seven trivial properties on a deliberately debiased set is well clear of chance.

A comparison we cannot make. No published Vina number exists for the AVE-debiased set. Comparing our AVE descriptor result against a full-set docking result would not be like for like and we do not do so. Vina would presumably also fall under debiasing; by how much is unknown to us.

3.4 A co-folding model exceeds the floor in combination — but only the standalone score is deployable

	Mpro	Factor Xa
descriptors	0.7654	0.7041
Boltz-2 alone	0.7913	0.7227
descriptors + Boltz-2	**0.8588**	**0.7791**
Boltz-2 − descriptors	+0.026 [−0.023, +0.074]	+0.019 [−0.030, +0.066]
combination − descriptors	**+0.093 [+0.062, +0.125]**	**+0.075 [+0.051, +0.099]**

Both halves replicate across a viral protease and a coagulation factor with independently assembled panels: the standalone comparison fails on both, the combination clears on both with overlapping intervals.

Both rows are fitted on each target's own labels. The combination in the table above is an out-of-fold logistic regression over the target's actives and inactives, and so carries exactly the supervision caveat §3.3 attaches to the descriptor baseline. On a target with no known actives there is nothing to fit it with. We tested whether the gain survives when the weights come from a different target — fitting on one panel in full and applying the model cold to the other, in both directions.

Boltz-2 is not less property-driven than docking — descriptors explain 63.1% of its score and 62.8% of Vina's. The difference is the residual. Regressing the seven descriptors out of each score out-of-fold and ranking actives on the remainder:

score	raw AUROC	residual AUROC	12-seed range
Vina, repaired receptor (Mpro)	0.4530	0.5187	[0.5131, 0.5259]
Vina, raw receptor (Mpro)	0.4079	0.5092	[0.4980, 0.5222]
gnina Vina term (Mpro)	0.3791	0.4941	[0.4887, 0.5017]
gnina CNNaffinity (Mpro)	0.5389	0.4994	[0.4908, 0.5091]
gnina CNNscore (Mpro)	0.5041	0.5426	[0.5320, 0.5592]
Vina, repaired receptor (Factor Xa)	0.6775	0.5733	[0.5634, 0.5810]
Boltz-2 `prob_binary` (Mpro)	0.7913	**0.6567**	**[0.6439, 0.6648]**

Six docking scores fall between 0.494 and 0.573; Boltz-2 sits at 0.657. The residual estimate moves with the cross-validation fold assignment, so each value is a mean over twelve fold seeds with its full range given, and the separation holds at the worst case — the highest docking residual observed across all seeds is 0.581, the lowest Boltz-2 residual 0.644. It also survives the choice of estimator: under linear rather than gradient-boosted residualisation the docking band is 0.418–0.614 and Boltz-2 is 0.748. The residual analysis is single-target — we did not compute a Factor Xa Boltz-2 residual.

The increment transfers; the model does not. Cold, the Boltz-2 increment clears the floor in both directions (+0.088 [+0.056, +0.119] Mpro → Factor Xa; +0.248 [+0.210, +0.287] the other way), but the transferred *models* are largely unusable. Sorted by what a screener actually has:

configuration	Mpro	Factor Xa	labels required
Boltz-2 alone, rank by score	**0.7913**	**0.7227**	**none**
descriptors, in-target out-of-fold	0.7673	0.6994	the target's own actives
descriptors + Boltz-2, in-target out-of-fold	0.8647	0.7803	the target's own actives
descriptors, cold from the other target	0.4553	0.4011	another target's actives
descriptors + Boltz-2, cold from the other target	0.7036	0.4890	another target's actives

Two things follow. **Seven descriptors fit on one target and applied to another rank actives below chance in both directions** — inverted transfer, not weak transfer. The property signature separating actives on one panel is close to the opposite of the other's, which is the strongest evidence here that the descriptor baseline measures target-specific bias rather than chemistry that generalises, and is why §3.3 treats it as a bias control.

And on a novel target the only usable configuration is the standalone score, which needs no labels, no receptor preparation and no docking box. That is the configuration reported as *not demonstrated* at the top of this section. Both readings are correct and they answer different questions: the in-target comparison asks whether Boltz-2 beats a baseline fitted on the answers; the prospective question asks what can be run when there are no answers to fit to. Under the first it fails; under the second it is the only candidate that works.

Residualising against ligand descriptors is a weaker test than receptor ablation. Binding free energy genuinely covaries with size and lipophilicity, so a physically correct scorer would also lose signal under this operation — that is why ligand efficiency exists. The defensible claim is narrow: the part of these docking scores orthogonal to seven descriptors does not rank actives in these sets, and on these two targets the ceiling appears to be a property of the scoring approach rather than of the task. A third scoring approach clearing it would be the test that generalises this.

4. Limitations

Two self-assembled targets for the six comparisons, and one of them (Mpro) is a target where our docking protocol scores below chance. LIT-PCBA broadens the descriptor result to 15 targets but not the comparisons, which remain a single-benchmark measurement.

Three of the six arms sit on the donor-defective receptor, as set out in §3.2 and Supplementary Table S2. The pose-ensemble arm has since been re-run on the repaired receptor and its verdict changed; whether the DiffDock null survives the same treatment is untested, and that null should be read as conditional.

Four of the six pre-registered bars predate the floor correction. Three of them cite a single "≈0.04 measurement floor" that §3.1 shows was conflating two quantities. The bars stand as written and the verdicts in §3.2 are read against them; the corrected floors are reported beside those verdicts, never substituted for them.

Cross-validation fold assignment moves AUROC differences on these panels by 0.01–0.03, which is the same size as the effects being measured. Two of our own results were misled by single-seed estimates: the residual band in §3.4, and the pose-ensemble arm, where a published refutation proved to be a 2nd-percentile draw. Every difference of this magnitude is reported here as a distribution over fold seeds with its range, and we would treat any single-seed AUROC difference below about 0.05 on a benchmark of this size as uninterpretable.

One analysis script was never committed. The original pose-ensemble implementation exists only as its pre-registration and its write-up, so its published number could not be re-derived and had to be reconstructed from a prose description. Every script behind the results reported here is in the repository with its outputs.

The resolution floors are measured on one benchmark of one size, on the unprotonated receptor. They should be re-measured rather than assumed elsewhere; the method costs one extra scoring pass. An earlier version of this analysis reported a single 0.039 floor and described it as a comparison of identical poses; it is not, and the two variants are separated here.

The descriptor baselines are supervised and docking is not, as stated in §3.3. This is the limitation most likely to be mistaken for a stronger claim than we make.

No Vina number on AVE-debiased LIT-PCBA, and only one source for the full-set figure.

The Boltz-2 evaluation is contested territory. A 2025 and a 2026 paper disagree about its virtual screening performance [6,7]. Our contribution is the pre-registration and the residual decomposition, not the discovery that it works.

A caveat we withdrew. Boltz-2's first run was blind while its docking comparator was given the binding site, and we recorded that as a handicap making +0.093 a lower bound. Supplying the pocket produced 0.7783 against the blind run's 0.7913, with the two runs correlating at $r = 0.952$. The caveat was unnecessary and is withdrawn. It is reported here because a pre-registered caveat that turns out to be wrong is part of the record.

5. Data and code availability

All pre-registration files, amendments, analysis code and raw outputs are public at **https://github.com/AegisMindApp/screening-decomposition**, with the commit hash and date for each pre-registration given in Supplementary Table S1. The history is intact rather than a snapshot, so any cited commit can be resolved and dated directly (`git log -1 --format=%ad --date=iso <commit>`).

The dates are git metadata carried through from the repository this was exported from; they are not a third-party timestamp, and a reader can confirm ordering and internal consistency but cannot from this alone exclude authored metadata. The resolution-floor analysis is reproducible from that repository with no dependencies beyond the Python standard library, and asserts a positive control against previously published values before reporting; that control failed once, on 6 September 2026, and the defect it caught is recorded in the repository alongside the result.

Two comparisons — the search-effort arm and the LIT-PCBA descriptor baseline — were not pre-registered and are identified as such in S1.

References

[1] Chen L, Cruz A, Ramsey S, Dickson CJ, et al. (2019) Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening. *PLOS ONE* 14:e0220113. doi:10.1371/journal.pone.0220113 [2] Sieg J, Flachsenberg F, Rarey M (2019) In Need of Bias Control: Evaluating Chemical Data for Machine Learning in Structure-Based Virtual Screening. *J Chem Inf Model* 59:947–961. doi:10.1021/acs.jcim.8b00712 [3] Tran-Nguyen VK, Jacquemard C, Rognan D (2020) LIT-PCBA: An Unbiased Data Set for Machine Learning and Virtual Screening. *J Chem Inf Model* 60:4263–4273. doi:10.1021/acs.jcim.0c00155 [4] Abo-Dahab Y, Xiang X, Chun J, Zhao L (2026) Benchmarking Single-Pose Docking, Consensus Rescoring, and Supervised ML on the LIT-PCBA Library: A Critical Evaluation of DiffDock, AutoDock-GPU, GNINA, and DiffDock-NMDN. arXiv:2605.01681 [5] Sunseri J, Koes DR (2021) Virtual Screening with Gnina 1.0. *Molecules* 26(23):7369. doi:10.3390/molecules26237369 [6] Furui K, Ohue M (2025) Boltzina: Efficient and Accurate Virtual Screening via Docking-Guided Binding Prediction with Boltz-2. arXiv:2508.17555 [7] Wan S, Zhang X, Xue X, Coveney PV (2026) On the Reliability of AI Methods in Drug Discovery: Evaluation of Boltz-2 for Structure and Binding Affinity Prediction. arXiv:2603.05532 [8] Huang A, Knight IS, Naprienko S (2025) Data Leakage and Redundancy in the LIT-PCBA Benchmark. arXiv:2507.21404

Supplementary material

S1. Pre-registration index. Every reading rule — endpoint, threshold, control, abort condition — was committed before the corresponding data existed. Each row gives the file, the commit that introduced it, and that commit's date. Amendments are appended to the file rather than editing the original text, with the reason recorded in place.

Paths are relative to `analysis/`; the file is `PREREGISTRATION.md` except where named.

arm	directory	public	origin	committed
Scoring function (gnina rescore)	`three_arm_docking/`	`f5e42306a`	`cd8f55639`	2026-08-31 08:46
Receptor preparation repair	`receptor_prep/`	`05d9f78ec`	`1f68b20c4`	2026-08-31 14:26
Pose ensemble vs top pose	`pose_ensemble/`	`d5d08c1ac`	`448d954e5`	2026-09-02 22:52
Pose generator (DiffDock-L)	`three_arm_docking/`	`f5e42306a`	`cd8f55639`	2026-08-31 08:46
Boltz-2, Mpro, arm 1 (blind)	`boltz2/`	`8df29ba21`	`9e3083d02`	2026-09-03 08:27
Boltz-2 analysis script, written before the data	`boltz2/analyse_boltz2.py`	`42f94969b`	`5afb5bec8`	2026-09-03 13:51
Boltz-2, Mpro, arm 2 (pocket-conditioned)	`boltz2/PREREGISTRATION_ARM2.md`	`064969caf`	`e96d4fea1`	2026-09-04 11:28
Boltz-2, Factor Xa	`boltz2/PREREGISTRATION_TARGETS.md`	`00baa6db9`	`68bd9b94a`	2026-09-04 11:40

The public repository was produced from the origin repository with `git filter-repo`, which rewrites commit hashes but preserves author and committer dates. Both hashes are given so a reader can check either. Verify a date with `git log -1 --format=%ad --date=iso <commit>`.

Commit dates above are git metadata carried through from the project's own history; they are not a third-party timestamp, and a reader should treat them as the authors' record rather than as independent verification of ordering.

Bars citing a "≈0.04 measurement floor" — the pose-ensemble arm and both Boltz-2 arms — were fixed before the floor correction described in §3.1 and S3. They are read as written; see §3.2.

The search-effort comparison (exhaustiveness 4 vs 32) and the LIT-PCBA descriptor baseline were not separately pre-registered; both are reported here as such. The search-effort arm reuses two runs that pre-dated the pre-registration programme, and the LIT-PCBA baseline applies a method already fixed for the Mpro and Factor Xa panels to a third-party dataset.

S2. Per-arm experimental conditions. The six interventions are single-factor comparisons against differing baselines, not a partition of one pipeline. Four were run before the receptor defect described in §3.2 was found and repaired.

intervention	baseline AUROC	receptor	exhaustiveness	n	poses held fixed?
Scoring function (gnina CNN)	0.3992	unprotonated	4	743	**yes**

intervention	baseline AUROC	receptor	exhaustiveness	n	poses held fixed?
Receptor repair	0.4079	unprotonated → protonated	4	751	receptor coordinates fixed; ligands re-docked
Pose ensemble	—	unprotonated	4	751	**yes** (same run, 5 modes)
Pose generator (DiffDock-L)	—	unprotonated	4 (Vina arm)	751	no
Pocket conditioning (Boltz-2)	0.7913	none — co-folding from sequence	n/a	750	n/a
Eight-fold search effort	0.4079 (exh 4)	unprotonated	4 → 32	753	no

S3. Resolution floor, both variants. `analysis/docking_value/floor_interval.py`, `FLOOR_INTERVAL.json`. Paired bootstrap over compounds, 10,000 resamples, unprotonated receptor.

	value	95% CI	n	r between the two score vectors
Protocol floor — exhaustiveness-32 Vina cache vs gnina `--score_only` on the exhaustiveness-4 poses	0.0393	[+0.0235, +0.0557]	745	0.9372
Scoring floor — exhaustiveness-4 Vina cache vs gnina `--score_only` on the same exhaustiveness-4 poses	0.0201	[+0.0112, +0.0286]	743	0.9700

Neither score vector contains a compound differing from its counterpart by more than 2 kcal/mol. An earlier description of the protocol floor as a comparison of "identical poses" was incorrect and is corrected in `analysis/docking_value/FLOOR_CORRECTION.md`.

S4. LIT-PCBA per-target results. `analysis/litpcba/descriptor_baseline_full.json` and `descriptor_baseline_ave.json`.

target	full: actives	inactives	AUROC	AVE: actives	inactives	AUROC
ADRB2	17	25,000	0.7008	4	25,000	0.6016
ALDH1	7,168	25,000	0.6169	1,343	25,000	0.6123
ESR1_ago	13	5,583	0.7260	3	908	0.8737
ESR1_ant	102	4,948	0.7525	25	794	0.6497
FEN1	369	25,000	0.7596	91	25,000	0.7471
GBA	166	25,000	0.7456	41	25,000	0.7390
IDH1	39	25,000	0.7921	9	25,000	0.7838
KAT2A	194	25,000	0.6656	48	25,000	0.6032

target	full: actives	inactives	AUROC	AVE: actives	inactives	AUROC
MAPK1	308	25,000	0.7285	77	15,250	0.7129
MTORC1	97	25,000	0.5945	24	8,243	0.5686
OPRK1	24	25,000	0.8640	6	25,000	0.7672
PKM2	546	25,000	0.6743	136	25,000	0.6068
PPARG	27	5,211	0.8459	6	861	0.7195
TP53	79	4,168	0.5769	16	795	0.5050
VDR	884	25,000	0.6918	165	25,000	0.6510
median			**0.7260**			**0.6510**

S5. Panels. Mpro: 751 compounds, 257 active (34.2%), PDB 7VU6. Factor Xa: 886 compounds, 405 active (45.7%). The Factor Xa marginal-value comparison in §3.4 uses an earlier 854-compound freeze of the same panel; where a number from that freeze is quoted it is labelled.